

MEMS Accelerometers and Gyroscopes: A New Meaning to Motion Sensing

Imagine no buttons on your mobile handset. You flick the phone and the number is dialed. Your PDA falls from a height and still the data in the hard drive is protected. Your car responds to the skidding and stabilises itself... All these are possible with micro-machined motion sensors



Freescale MMA7660FC 3-axis digital accelerometer—Smart motion with power-select

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“Words represent your intellect. Gestures and movements represent your feelings.” That’s well said by Patricia Fripp, a renowned speech coach. Today, motion sensing has given a whole new face to user interface. It has added life to consumer electronics products, resulting in some smart and cool products and applications. It is widely used in

safety systems of vehicles due to the kind of accuracy and resolution it provides. Gesture controls are also used in mobiles and gaming devices. With the developments in micro-machining, the mechanical motion is sensed by electronic systems giving rise to intelligent control systems.

Micro-electro-mechanical systems (MEMS) are smart, intelligent micro-machines that sense the contextual motion and take suggested actions. MEMS

are microscopic structures integrated onto a silicon chip that combines mechanical elements with electronics. These micro-machines enable the chip to become intelligent. MEMS make it possible for systems of all kinds to become smaller, faster, more energy efficient and less expensive.

Interpreting the right type of motion

The success of motion sensing applications is driven by continuous innovation, creativity and development in micro-fabrication techniques. Motion sensing is a fairly old technology that was first used in aircraft navigation systems. With motion sensors currently taking the shape of micro-machines, it has captured the consumer market, finding its way into products like mobiles, cameras, gaming consoles, security systems and many more.

There are five types of motions that can be detected with a variety of accelerometers and gyroscopes which can be innovatively used in a wide range of applications. These are acceleration, vibration, shock, tilt and rotation. The first four types display some kind of linearity and acceleration. However, rotation is handled a little differently. Vibration can be described as a periodic acceleration and deceleration as opposed to shock which is non-periodic, instantaneous acceleration. Rotational motion is detected by gyroscopes.

A product designer needs a thorough understanding of the type of motion when deciding which type of